

## **Failed to reproduce 600-qubit VQE because of being out of memory**

<https://github.com/tencent-quantum-lab/tensorcircuit/issues/61>

ROW 120

# Failed to reproduce 600-qubit VQE because of being out of memory #61

[New issue](#)

Closed



royess opened on Sep 16, 2022

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I tried to reproduce the 600-qubit VQE mentioned in [the whitepaper](#) by running a script slightly modified from [examples/vqe\\_extra\\_mpo.py](#). Modifications are:

1. Only do one forward evaluation;
2. Use CPU instead of GPU.

I use a machine with 36 cores and 251GB memory run the script. The backend is set as tensorflow. But the job is killed because of being out of memory, while the whitepaper shows that a 40GB GPU is enough for a same job.

► Version information.

► The script.



Assignees

No one assigned

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None yet

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No branches or pull requests

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Participants



refraction-ray on Sep 17, 2022

Contributor

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Is the out of memory error raised at jit staging time or runtime?

Also, one should set a rather large time buffer for the cotengra optimizer to search for the better tensor network contraction path, `max_time=360` is not enough. The path searching for the  $n=600$  problem may takes one hour or so to get an reasonable path. The way to check whether the path finder is good enough is focusing at the summary information of contraction path, the WRITE metric is equivalent to the memory consumption  $2^{\text{WRITE}}$  up to some constant overhead



royess on Sep 17, 2022

Contributor

Author

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Is the out of memory error raised at jit staging time or runtime?

In my script, I suppose jit is not enabled. So the out of memory error should be at runtime.

And thanks for your suggestions, I will have another try by setting a larger `max_time`.



royess on Sep 18, 2022

Contributor

Author

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The memory cost is reduced to ~30GB when I use a `max_time=3600` for finding the contraction path. Many thanks!



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royess closed this as **completed** on Sep 18, 2022